



# Levenberg–Marquardt supported mathematical modeling of Darcy–Forchheimer magnetized cross nanofluid flow with melting cylindrical surface and chemical processes using viscous dissipation

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## ABSTRACT

Many processes, such as the melting of permafrost, the solidification of magma, and the liquefying of frozen surface, are influenced by the melting process. The importance of understanding the melting procedures in fluid dynamics is established by freezing and melting of soil near the ground-based pipes' heat exchanger loops. This work aims to study the impact of melting on the mass and heat transport characteristics of a Darcy–Forchheimer porous media in Cross-Nano fluidic flow in the direction of a cylindrical surface that is horizontally extended at the stagnation point of MHD. The method of heat transport is elaborated by viscous dissipation. Using the Cross nanofluidic model, parameters of Thermophoresis and Brownian motion studied. Furthermore, their influences on the solutal field are observed by chemical collaborations. The nonlinear coupled PDEs monitoring the fluid flow are transmuted into nonlinear ODEs by executing the proper similarity transformation. By means of unconventional Levenberg–Marquardt Backpropagation Scheme (LMB-Scheme) pooled with Neural Networks of Artificial Intelligence (NNs of AI) equations being numerically solved. To assess the accuracy of the model under various conditions, the efficiency of the anticipated NNs of AI with LMB-Scheme is evaluated through several metrics of performance, including plots of Regression Analysis (RA), analysis of Error Histogram (EA), and Mean Squared Error (MSE). A detailed computational investigation is performed to analyze the effects of key governing parameters on the velocity, temperature, and concentration fields of the unsteady MHD Carreau nanofluid model. The physical formulation incorporates thermal radiation, variable viscosity, and chemically reactive species, and the governing nonlinear partial differential equations are transformed into a system of ordinary differential equations using appropriate similarity transformations. The resulting system is solved through an artificial neural network framework trained with the Levenberg–Marquardt backpropagation scheme, and its accuracy is assessed using standard performance metrics including mean squared error, regression analysis, and convergence evaluation. The revised abstract therefore concludes by clearly outlining the problem formulation, computational strategy, and validation methodology adopted in this study.

## Nomenclature

|        |                         |      |                |
|--------|-------------------------|------|----------------|
| $x, r$ | Cylindrical coordinates | $Ec$ | Eckert number  |
| $\rho$ | Density of Fluid        | $Sc$ | Schmidt number |

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(continued)

|          |                                  |       |                                |
|----------|----------------------------------|-------|--------------------------------|
| $\Gamma$ | Cross-time relaxation's constant | $c_s$ | Heat capacity of solid surface |
|----------|----------------------------------|-------|--------------------------------|

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|                                        |                                            |                  |                                          |
|----------------------------------------|--------------------------------------------|------------------|------------------------------------------|
| $c_p^*$                                | Specific heat at persistent pressure       | $N_b^*$          | Brownian motion                          |
| $\tilde{u}, \tilde{v}$                 | Components of velocity                     | $Pr$             | Prandtl number                           |
| $T_\infty$                             | Ambient temperature                        | $c_k$            | Drag coefficient                         |
| $\tilde{K}$                            | Rate of reaction                           | $N_r^*$          | Thermophoresis parameter                 |
| $\mu_0$                                | Zero shear rate viscosity                  | $T_0$            | Reference temperature                    |
| $B_0$                                  | Uniform magnetic field                     | $\tilde{D}_T$    | Coefficient of Thermophoresis diffusion  |
| $\mu$                                  | Viscosity of Fluid                         | $\tilde{\sigma}$ | Fluid's Electrical Conductivity          |
| $\tilde{\alpha}$                       | Thermal conductivity of fluid              | $\tilde{M}$      | Melting surface parameter                |
| $We$                                   | Weissenberg number                         | $C_\infty$       | Ambient concentration                    |
| $N^*$                                  | Parameter of chemical reaction             | $\tilde{F}$      | Forchheimer parameter                    |
| $\tilde{q}_w$                          | Sheer stress                               | $C_m$            | Surface concentration                    |
| $f$                                    | Dimensionless velocity                     | $C_w$            | Wall concentration                       |
| $l$                                    | Characteristics length                     | $a_0, d_0$       | Positive constants                       |
| $\tilde{\lambda}$                      | Fluid's Latent heat                        | $\tau$           | Ratio of heat capacity to the base fluid |
| $U_e(x)$                               | Free stream velocity                       | $\tilde{C}$      | Concentration                            |
| $T_w$                                  | Wall temperature                           | $\tilde{q}_s$    | Mass flux                                |
| $T_m$                                  | Surface temperature                        | $q$              | Stretching ratio parameter               |
| $\tilde{T}$                            | Temperature                                | $\beta$          | Parameter of Local curvature             |
| $\tilde{Nu}$                           | Nusselt number                             | $\theta$         | Dimensionless temperature                |
| $\tilde{Sh}_x$                         | Sherwood number                            | $w^*$            | Porosity parameter                       |
| $\phi$                                 | Dimensionless concentration                | $B^*$            | Magnetic field parameter                 |
| $\tilde{C}_f$                          | Local skin friction                        | $\tau_{rx}$      | Surface shear stress                     |
| $(Re)_x$                               | Local Reynolds                             | $R$              | Cylinder's radius                        |
| $\xi$                                  | Stream function                            | $\eta$           | Similarity variable                      |
| $k^*$                                  | Permeability of medium                     |                  |                                          |
| Terms for Nanofluid                    |                                            |                  |                                          |
| MHD                                    | Magneto-hydrodynamic                       | NF               | Nanofluid                                |
| CNFM                                   | Cross Nano-fluidic Model                   |                  |                                          |
| Terms for Artificial Intelligence (AI) |                                            |                  |                                          |
| NN                                     | Neural Networks                            | AI               | Artificial intelligence                  |
| RA                                     | Regression Analysis                        | FF               | Functions Fit                            |
| EH                                     | Error Histogram                            | TS               | Training State                           |
| PA                                     | Performance Analysis                       | EA               | Error Analysis                           |
| SC                                     | Solution Curves                            | MSE              | Mean-Squared Error                       |
| LMB-Scheme                             | Levenberg Marquardt Backpropagation Scheme | SNR              | Scenario                                 |

## 1. Introduction and motivation

Fundamental fluids with particles lesser than a nanometer, like kerosene, water, diol and oil are itemized as nanofluids. Since these particles are too tiny, we refer to them as nanoparticles. Nanoparticles can be found in metal, ceramics, carbon, metal nitrides, and a broad range of other materials. Basic fluids have been used as cooling agents in a diversity of technical and industrial applications. These have a lower heat conductivity than solids. Because of this, heat transfer through the dispersion of nanoparticles in ordinary fluids is extraordinarily efficient. Magnetohydrodynamics (MHD) plays a dynamic role in industry and many flow events. Aerodynamics, the automobile industry, biochemical industrial heat transfer, ceramic technology, metallurgical technology, mental operating procedures, and fluid dynamics are just a few of the interdisciplinary technical sectors where the MHD is helpful. The preferred route of the flow field is altered by MHD start due to irregular boundary layer configuration flow. Convection instruments like as nuclear reactor control systems, metallic casting, and material processing depend heavily on electromagnetic fields.

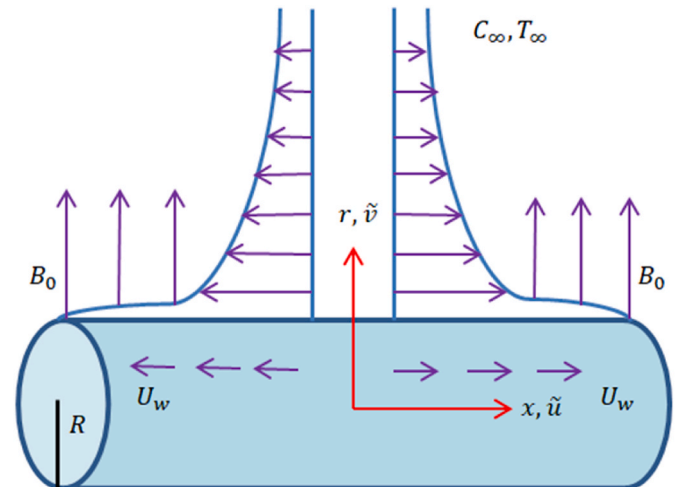
The concept of nanofluids was pioneered by Choi [1], who demonstrated that dispersing nanoparticles into conventional base fluids could markedly enhance thermal conductivity and open new pathways for advanced heat transfer applications. Building on this foundation, Yu and Choi [2] clarified the influence of nanoscale interfacial layers on thermal transport and later proposed a structured conductivity model based on a cubic distribution of spherical particles [3]. Li et al. [4] systematically discussed preparation techniques and characterization strategies for

nanofluids, while Heris et al. [5] experimentally confirmed improved convective heat transfer performance in duct flows using alumina–water suspensions. The role of magnetic fields and radiative effects in nanofluid systems was examined through two-phase and porous media frameworks by Sheikholeslami et al. [6] and Sheikholeslami et al. [7], whereas Vakili et al. [8] integrated experimental data with neural network modeling to predict radiative properties of carbon nanotube-based nanofluids. Convective MHD nanofluid behavior in porous enclosures was further investigated by Sheikholeslami and Shehzad [9]. Considerable attention has been devoted to activation energy and nonlinear rheological models: Khan et al. [10] analyzed stagnation-point flow of Cross nanofluid with chemical kinetics effects, Rasool et al. [11] explored Darcy–Forchheimer flow over stretching surfaces, and Khan et al. [12] evaluated three-dimensional Cross nanofluid flow with binary reactions and radiation. The combined impacts of electromagnetism forces and thermoelectric phenomena were addressed by Animasaun et al. [13], while Sabir et al. [14] contributed advanced nonlinear mathematical formulations applicable to transport modeling. A comprehensive synthesis of nanofluid heat transfer in macro- and micro-channels was provided by Liang and Mudawar [15]. Expanding and contracting cylinder configurations for Cross magneto-nanofluid flow were detailed by Ali et al. [16], and melting heat transfer characteristics were studied by Ali et al. [17], followed by stratification and activation energy considerations in three-dimensional configurations by Ali et al. [18]. Bioconvection and microorganism-induced transport in MHD nanofluids were examined by Kotha et al. [19], while Rasool et al. [20] numerically assessed Darcy–Forchheimer convective flow under magnetic influence. Broader reviews of nanofluid applications and thermohydraulic performance were presented by Li et al. [21] and Awais et al. [22]. The theoretical underpinnings of magnetohydrodynamic modeling were clarified by Pontin and Priest [23], supporting later analyses such as the magnetized radiative Sutterby nanofluid flow studied by Khan et al. [24] and the unsteady boundary layer behavior with Brownian and thermophoretic effects described by Kalpana et al. [25]. Radiative and chemically reactive MHD nanofluid transport over moving surfaces was analyzed by Arulmozhi et al. [26], and second-order slip with activation energy in cylindrical Cross nanofluid

**Table 1**

Comparison for  $-\theta'(0)$  with published results when  $\beta = N^*$ ,  $b = N^*$ ,  $t = Ec = \tilde{M} = We = 0$  for two different values of  $Pr$ .

| $Pr$ | Giann et al. [56] | Shahzad et al. [56] | Present results |
|------|-------------------|---------------------|-----------------|
| 1    | 0.58201           | 0.5822              | 0.5821          |
| 10   | 2.3080            | 2.3080              | 2.3083          |



**Fig. 1.** CNFM geometry with flow assumptions.

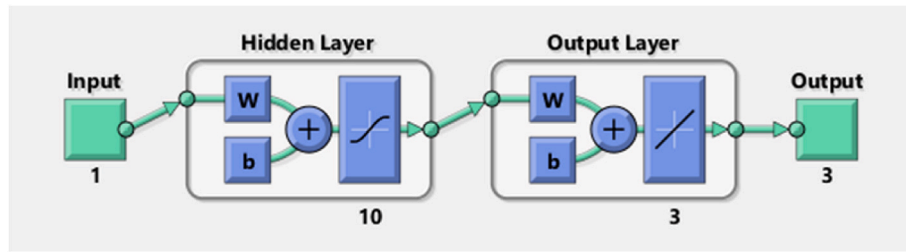


Fig. 2. Neural Network architecture showing input, activation function and output.

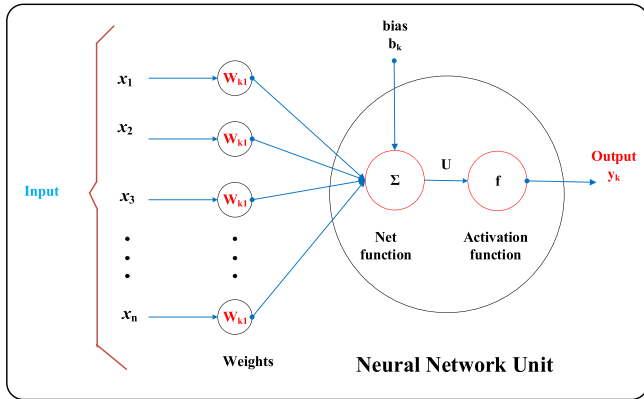


Fig. 3. A construction of a Single Neural model with input weights and activation function.

**Table 2**  
Statistical overview of the parameter values influencing the Velocity, Temperature and Concentration functions.

| Fun                                | Snrs | Cases | Parameters of Variables |        |      |      |
|------------------------------------|------|-------|-------------------------|--------|------|------|
|                                    |      |       | $q$                     | $Nb^*$ | $Pr$ | $Sc$ |
| Velocity Profile $f'(\eta)$        | 1    | 1     | 0.1                     | 0.1    | 4.0  | 0.1  |
|                                    |      | 2     | 0.15                    | 0.1    | 4.0  | 0.1  |
|                                    |      | 3     | 0.2                     | 0.1    | 4.0  | 0.1  |
|                                    | 2    | 1     | 0.1                     | 0.1    | 0.03 | 0.1  |
|                                    |      | 2     | 0.1                     | 0.1    | 0.06 | 0.1  |
|                                    |      | 3     | 0.1                     | 0.1    | 0.09 | 0.1  |
| Temperature Profile $\theta(\eta)$ | 3    | 1     | 0.1                     | 0.1    | 4.0  | 0.1  |
|                                    |      | 2     | 0.1                     | 1.1    | 4.0  | 0.1  |
|                                    |      | 3     | 0.1                     | 2.1    | 4.0  | 0.1  |
|                                    | 4    | 1     | 0.1                     | 0.1    | 4.0  | 0.1  |
|                                    |      | 2     | 0.1                     | 0.1    | 5.0  | 0.1  |
|                                    |      | 3     | 0.1                     | 0.1    | 6.0  | 0.1  |
| Concentration Profile $\phi(\eta)$ | 5    | 1     | 0.1                     | 0.1    | 4.0  | 0.1  |
|                                    |      | 2     | 0.1                     | 0.1    | 4.0  | 1.1  |
|                                    |      | 3     | 0.1                     | 0.1    | 4.0  | 2.1  |
|                                    | 6    | 1     | 0.1                     | 0.1    | 4.0  | 0.1  |
|                                    |      | 2     | 0.1                     | 0.1    | 5.0  | 0.1  |
|                                    |      | 3     | 0.1                     | 0.1    | 6.0  | 0.1  |

**Table 3**  
Values of MSE and Levenberg-Marquardt Scheme output data.

| Snr | MSE data             |                      |                      | Performance Grids    | Gradient Grids       | Mu Grids             | Closing Epoch | T/Sec |
|-----|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|---------------|-------|
|     | Training             | Validation           | Testing              |                      |                      |                      |               |       |
| 1   | 2.77E <sup>-9</sup>  | 5.42E <sup>-9</sup>  | 4.67E <sup>-9</sup>  | 2.77E <sup>-9</sup>  | 9.98E <sup>-08</sup> | 1.00E <sup>-08</sup> | 268           | 03    |
| 2   | 1.67E <sup>-9</sup>  | 2.59E <sup>-9</sup>  | 2.31E <sup>-9</sup>  | 1.68E <sup>-9</sup>  | 9.81E <sup>-08</sup> | 1.00E <sup>-8</sup>  | 178           | 02    |
| 3   | 1.84E <sup>-10</sup> | 1.86E <sup>-10</sup> | 1.21E <sup>-10</sup> | 1.85E <sup>-10</sup> | 8.99E <sup>-08</sup> | 1.00E <sup>-9</sup>  | 753           | 08    |
| 4   | 1.88E <sup>-9</sup>  | 3.07E <sup>-9</sup>  | 2.31E <sup>-9</sup>  | 1.89E <sup>-9</sup>  | 9.91E <sup>-08</sup> | 1.00E <sup>-8</sup>  | 238           | 02    |
| 5   | 2.13E <sup>-10</sup> | 4.40E <sup>-10</sup> | 5.36E <sup>-10</sup> | 2.14E <sup>-10</sup> | 9.98E <sup>-08</sup> | 1.00E <sup>-9</sup>  | 355           | 04    |
| 6   | 1.56E <sup>-9</sup>  | 3.84E <sup>-9</sup>  | 2.19E <sup>-9</sup>  | 1.57E <sup>-9</sup>  | 9.94E <sup>-08</sup> | 1.00E <sup>-8</sup>  | 296           | 03    |

bioconvection was investigated by Aljaloud et al. [27]. Theoretical modeling of Sutterby nanofluids with internal heat generation or absorption was further advanced by Irfan et al. [28] and reiterated in related analysis by Irfan et al. [29]. Marangoni-driven Cross nanofluid flow with convective boundary conditions was numerically examined by Abbas and Khan [30], and similar porous medium Marangoni configurations involving hybrid nanofluids were explored by Qureshi and Altmeier [31]. The influence of quadratic radiation and activation energy in oblique stagnation hybrid nanofluid flow over cylindrical geometries was reported by Madhu et al. [32], while Zia et al. [33] incorporated artificial intelligence techniques to assess thermally magnetized EMHD Cross hybrid nanofluids. The contribution of microorganisms to radiative nanofluid heat transport along cylindrical surfaces was highlighted by Arshad et al. [34], and analytical solution methodologies for damped telegraph-type transport equations relevant to fluid modeling were provided by Qureshi et al. [35]. Most recently, Iqbal and Akbar [36] implemented deep learning tools for thermodynamic and entropy generation analysis in nanofluid systems, demonstrating the growing integration of computational intelligence with advanced heat transfer research.

AI of Neural Networks have revealed its significance in a variety of engineering and chemical applications, particularly when joined with LMB-Scheme. NNs of AI can clutch hard and nonlinear problems like mass and heat conveyance. They deal uncommon to plain numerical and analytical approaches. Due to resembling with the learning process of human's brain, AI of NNs is far more effective in gathering challenging correspondences across datasets and producing noteworthy prediction capabilities. For a variety of applications, including the examination of bio-convection flow in nanofluids and assessing the efficacy of heat transfer in solar collectors with irregular heat sources, NNs of AI perform far better when combined with LMB Algorithm. Researchers used this strategy for the examination of the robotic exploiters by employing techniques of NNs of AI to tackle inverse and forward kinematic problems. Moreover, by performing research for a wide range of chemical and engineering applications, NNs of Artificial Intelligence have shown its utility and conformance across various disciplines. In addition, current progresses in the application of NN systems in unification with numerical-based swarming/evolutionary experiential methods for stochastic computing have stretched their effectiveness, mainly in modeling complex systems other than conventional ODEs, opening the door for more investigation in engineering modeling and simulation. Recent investigations increasingly integrate artificial intelligence,

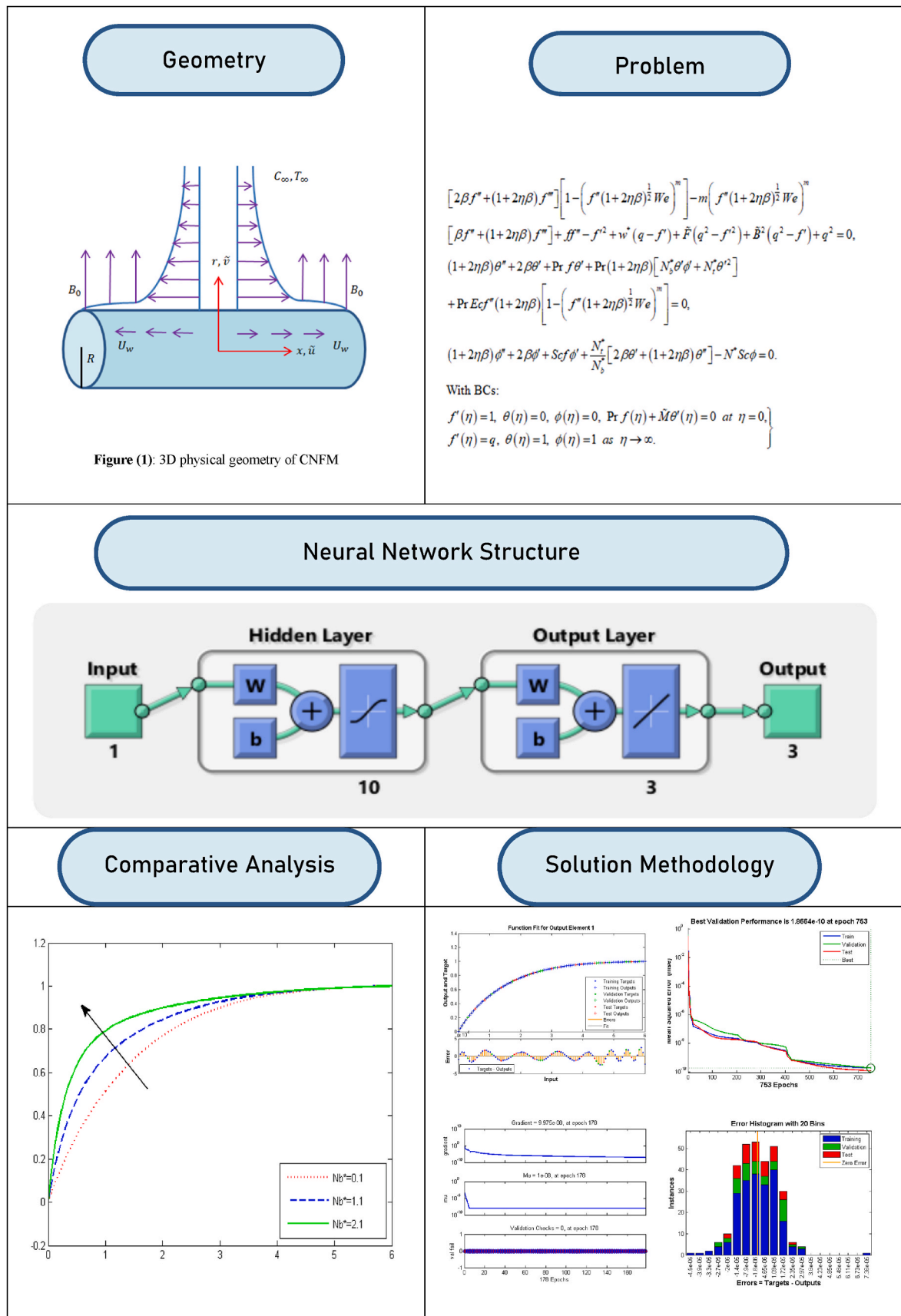
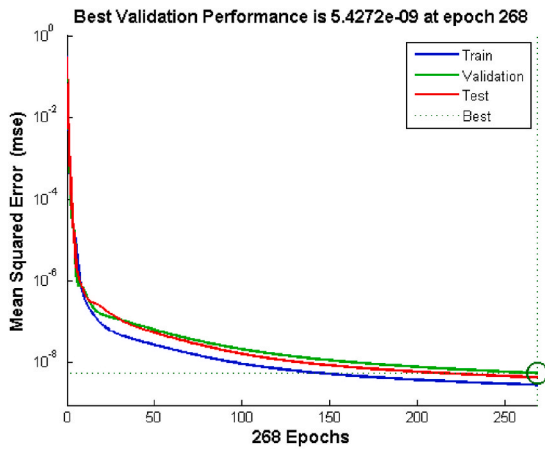
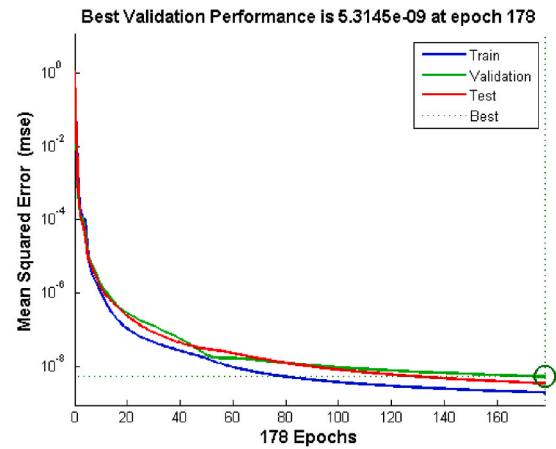


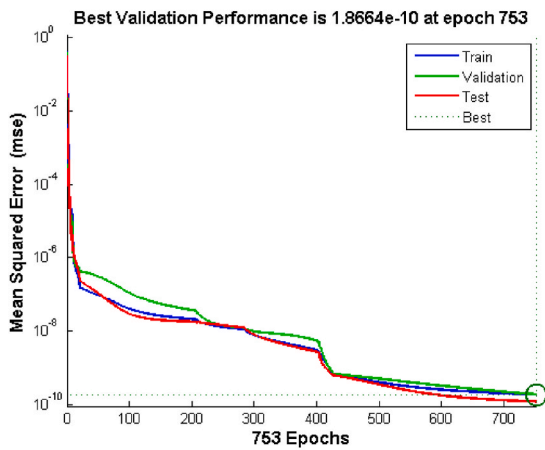
Fig. 4. Flow chart representing flow process of CNFM.



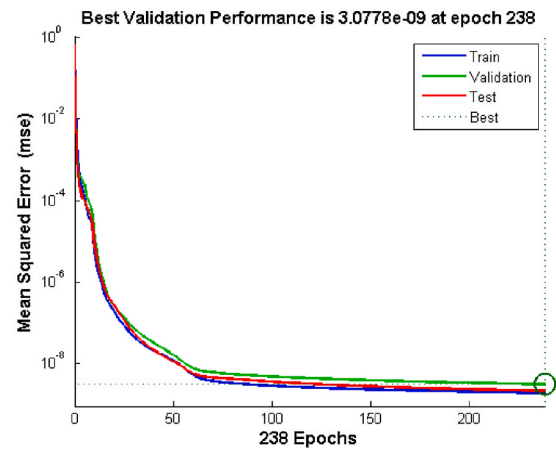
(a) Performance Validation output for 1st Snr



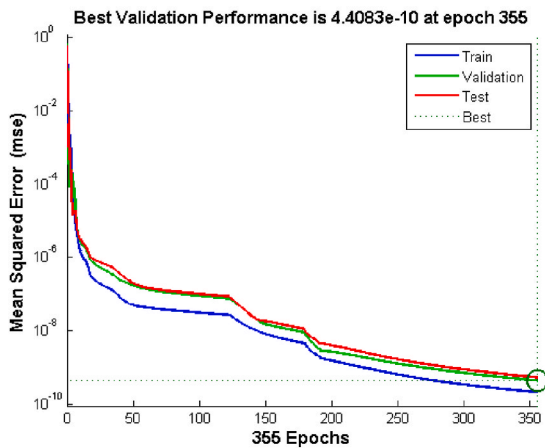
(b) Performance Validation output for 2nd Snr



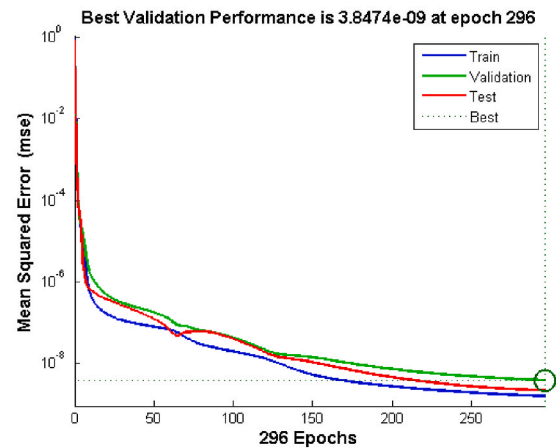
(c) Performance Validation output for 3rd Snr



(d) Performance Validation output for 4th Snr



(e) Performance Validation output for 5th Snr

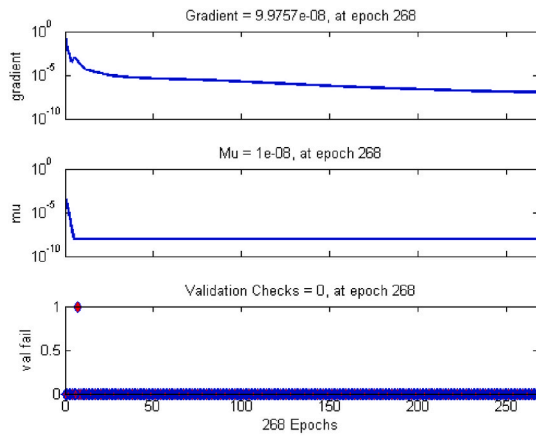


(f) Performance Validation output for 6th Snr

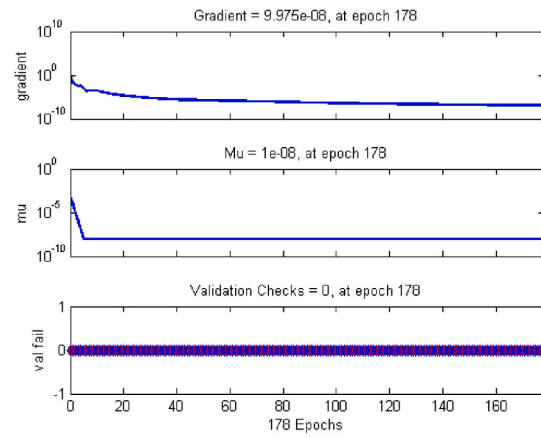
Fig. 5. Performance Validation outputs of CNFM.

particularly Levenberg–Marquardt (LM) and related neural network algorithms, into the modeling of complex nanofluid systems. Ali et al. [37] applied an LM-based neural framework to examine MHD Williamson nanofluid flow incorporating radiation, Joule heating, and

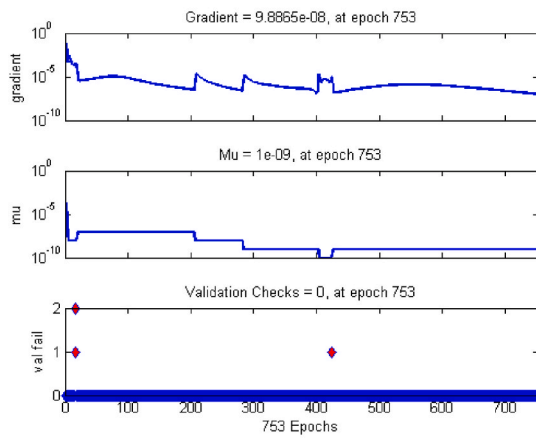
chemical reaction effects. Esfe et al. [38] optimized neural network structures to accurately predict the dynamic viscosity of oil-based hybrid nanofluids containing MgO and MWCNT nanoparticles. Sundar and Mouli [39] combined experimental measurements with LM-trained



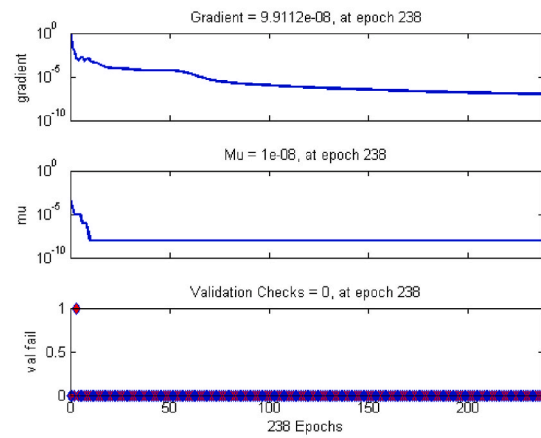
(a) Outcomes of TS for 1st Scenario



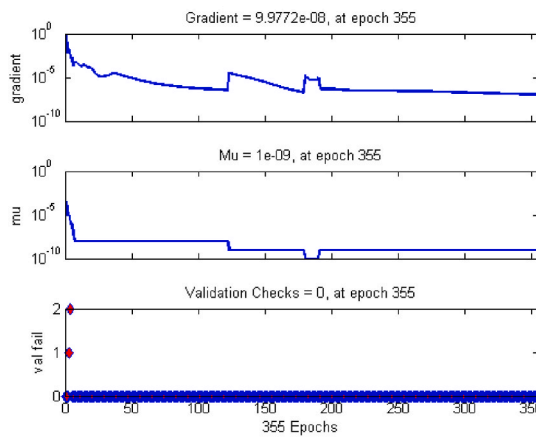
(b) Outcomes of TS for 2nd Scenario



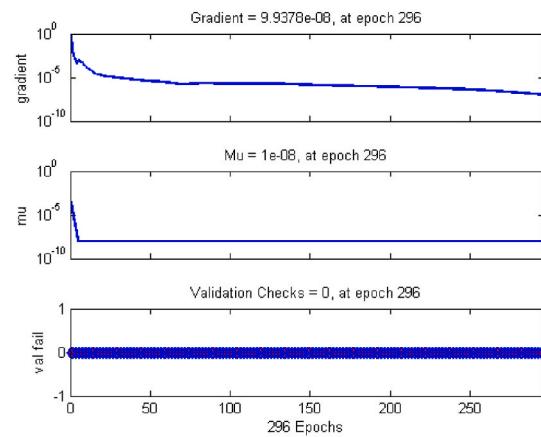
(c) Outcomes of TS for 3rd Scenario



(d) Outcomes of TS for 4th Scenario



(e) Outcomes of TS for 5th Scenario

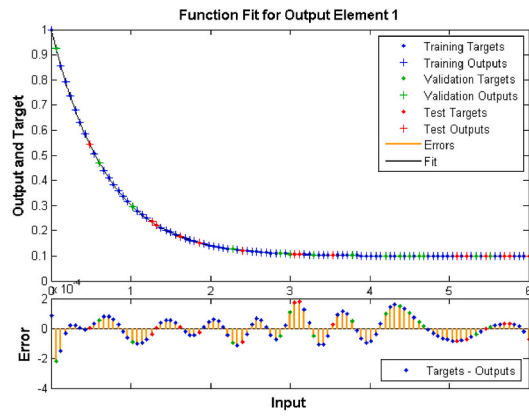


(f) Outcomes of TS for 6th Scenario

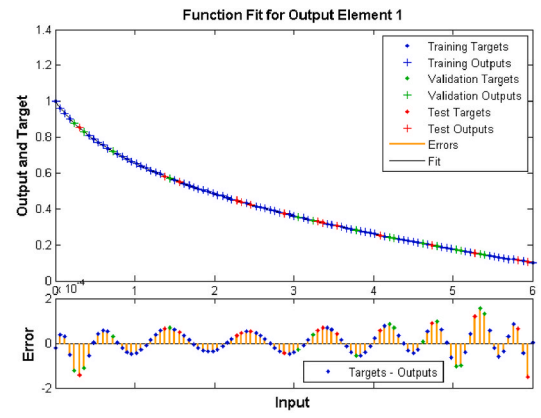
Fig. 6. TS aspects of LMB-Scheme for CNFM.

neural models to forecast thermal and hydraulic performance of MgO/water nanofluids in thermosyphon collectors, while Sharma et al. [40] utilized backpropagation neural computation for ternary nanofluid transport between parallel plates under specialized boundary constraints. Intelligent prediction of carbon nanotube–water nanofluid

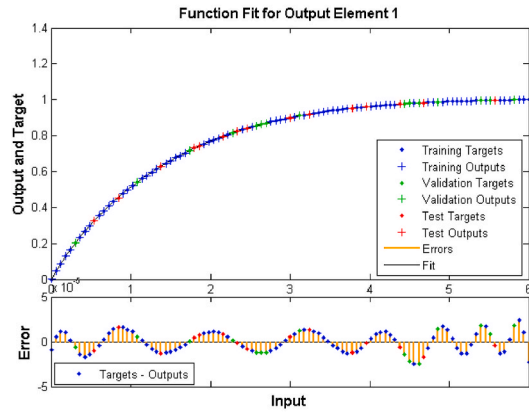
behavior between rotating disks was conducted by Ali et al. [41], and Almeida et al. [42] implemented LM-based artificial neural networks to simulate Cross–Carreau nanofluid flow over curved stretching geometries. Shah et al. [43] extended ANN methodologies to characterize bioconvective nanofluid dynamics, whereas Akbar et al. [44] simulated



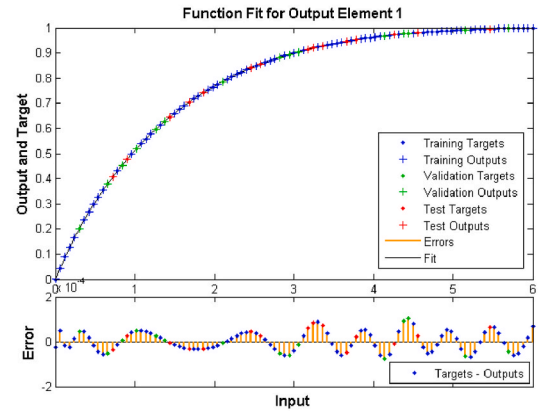
(a) FF for 1st Scenario



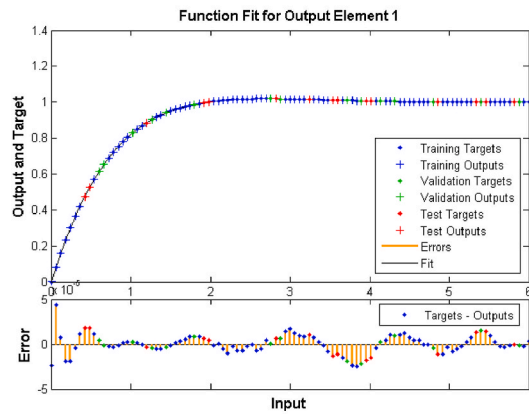
(b) FF for 2nd Scenario



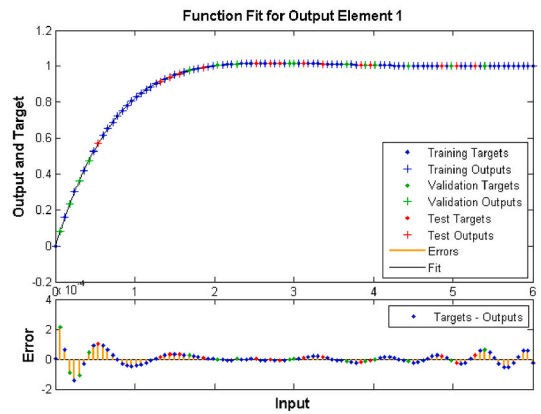
(c) FF for 3rd Scenario



(d) FF for 4th Scenario

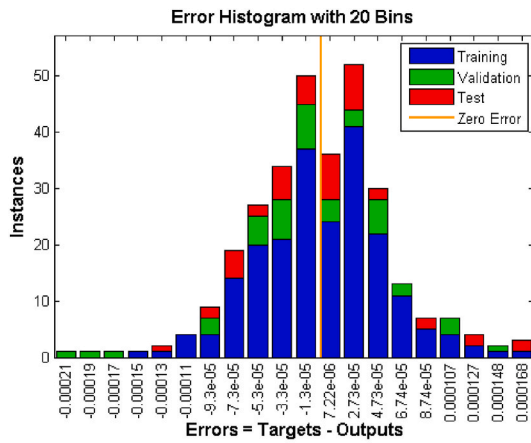


(e) FF for 5th Scenario

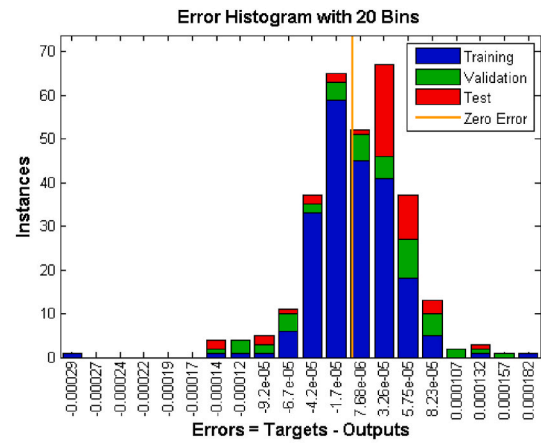


(f) FF for 6th Scenario

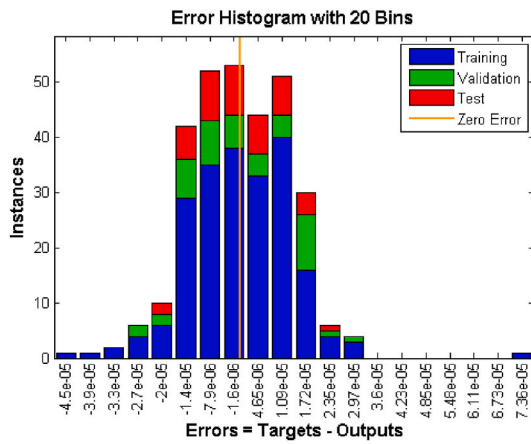
Fig. 7. Examination of LMB-Scheme results with relevant solution of CNFM.



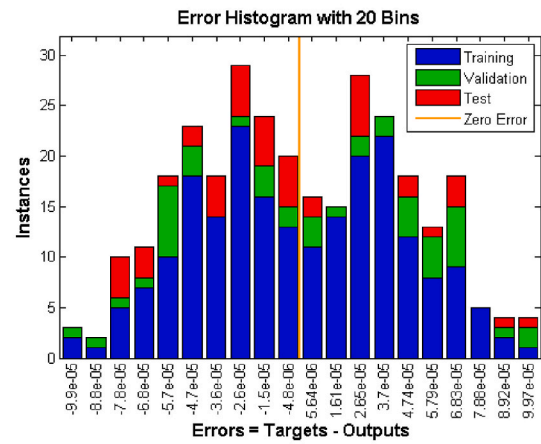
(a) AEH for 1st Scenario



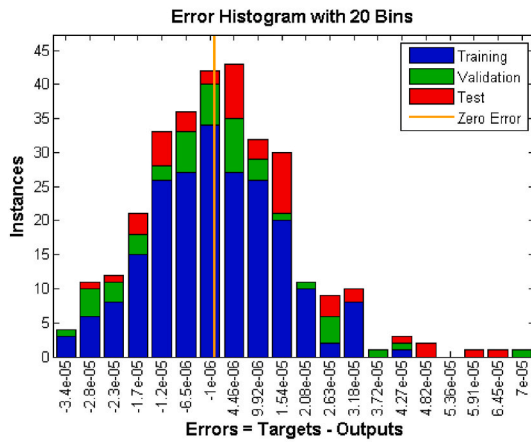
(b) AEH for 2nd Scenario



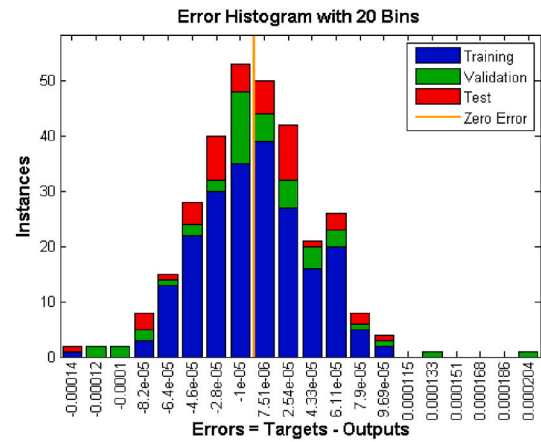
(c) AEH for 3rd Scenario



(d) AEH for 4th Scenario



(e) AEH for 5th Scenario



(f) AEH for 6th Scenario

Fig. 8. EH for LMB-Scheme result of CNFM.

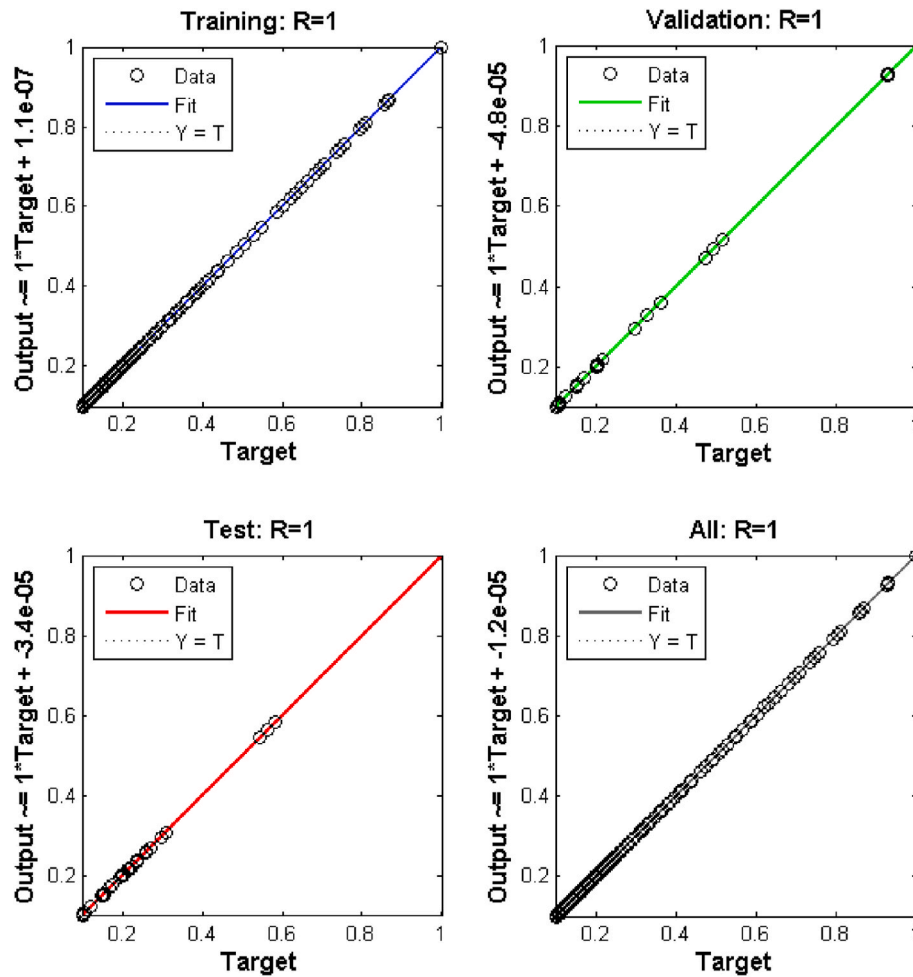


Fig. 9. RA for LMB-Scheme of 1st snr of CNFM.

hybrid boiling nanofluid transport through porous stretching sheets using neural optimization techniques. Khan et al. [45] addressed thermophoretic particle deposition in three-dimensional Cross nanofluid flow via ANN modeling, and Sakkaravarthi et al. [46] focused on entropy minimization in rotating disk configurations with nonlinear radiation effects. Yaseen et al. [47] employed LM-trained networks for hybrid nanofluid heat transfer between rotating disks, and Kumar et al. [48] comparatively assessed entropy optimization in Cross hybrid nanofluid systems. Faisal et al. [49] analyzed nanolayer morphology influences on magnetized tri-hybrid nanofluid flow through neural optimization, while Qureshi et al. [50] proposed a supervised machine learning paradigm to quantify melting and dissipative mechanisms in Darcy–Forchheimer ternary hybrid nanofluid flows. Expanding machine learning beyond fluid mechanics, Qureshi et al. [51] investigated biomedical data analytics, and Qureshi et al. [52] developed ANN models for solar-energy-assisted Cross fluid systems. Arshad et al. [53] designed AI-based predictive neural architectures for chemically reactive magnetized Darcy–Forchheimer nanofluid flow, and Akkilic et al. [54] demonstrated Bayesian-regularized deep networks for epidemiological modeling, highlighting methodological parallels. Iqbal et al. [55] incorporated ANN techniques into MHD peristaltic transport of non-Newtonian fluids, and subsequent works by Iqbal and collaborators [56–62] integrated statistical modeling, neural networks, Bayesian regularization, and sensitivity analysis to optimize hybrid nanofluid thermal systems, solar-powered aircraft collectors, and bioconvective peristaltic mechanisms in porous media. A comprehensive review of ANN strategies for magnetohydrodynamic hybrid nanofluid flows was presented by Abdal et al. [63], while Baithalu et al. [64] applied LM

neural computation to micropolar nanofluid transport with non-Fourier heat flux models. Wang et al. [65] utilized AI-integrated LM estimation for mixed convection tri-hybrid nanofluid systems with chemical reactions, and Ramesh et al. [66] examined thermodynamic activity in ternary nanofluid flow across permeable surfaces. Shao et al. [67] adopted stochastic AI approaches to evaluate radiative magnetized Cross fluid flow with variable properties, and Arshad et al. [68] implemented intelligent algorithms to study chemically reactive tetra-hybrid nanofluids in biomedical-inspired stenotic artery configurations. Machine learning-based predictive analysis of Falkner–Skan Cross nanofluid flow was conducted by Arshad et al. [69], while Abbasi et al. [70] employed artificial intelligence techniques to assess nonlinear radiative Williamson nanofluid flow incorporating motile microorganisms. Collectively, these contributions demonstrate the accelerating convergence of advanced nanofluid transport modeling with neural network-driven intelligent computation for enhanced accuracy, optimization, and multidisciplinary applications.

### 1.1. Research gap and motivation

Despite notable advancements in artificial intelligence-assisted solvers for nonlinear fluid mechanics, significant gaps remain in the application of Levenberg–Marquardt backpropagation (LMB) neural network schemes to complex rheological models such as the unsteady three-dimensional Carreau nanofluid model (CNFM). Existing studies largely emphasize steady or two-dimensional configurations, simplified nanofluid assumptions, or isolated effects of chemical reaction without simultaneously incorporating activation energy within a unified AI-

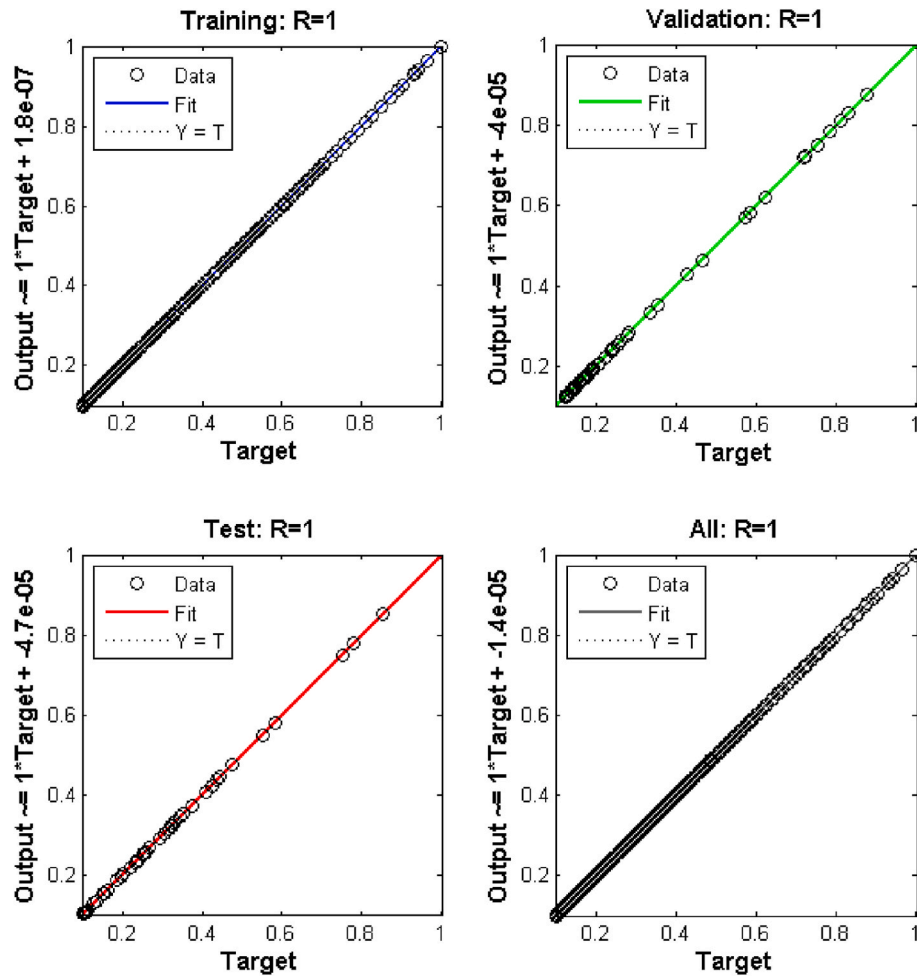


Fig. 10. RA for LMB-Scheme of 2nd snr of CNFM.

based computational framework. Moreover, although similarity transformations are widely used to convert governing nonlinear partial differential equations (PDEs) into ordinary differential equations (ODEs), the integration of such transformed systems into stochastic AI-built neural networks with structured dataset engineering, systematic training-validation-testing procedures, and rigorous statistical authentication remains underdeveloped. In particular, gaps can be identified as follows:

**Modeling Gap:** Limited AI-driven treatment of unsteady three-dimensional CNFM incorporating movement control and fluid motion assumptions under realistic boundary conditions.

**Computational Intelligence Gap:** Lack of structured datasets generated via advanced numerical strategies such as the Adam technique for robust LMB-Scheme training.

**Validation Gap:** Minimal use of comprehensive AI performance diagnostics including mean squared error (MSE), regression analysis, training state functions, fitness functions, and error histogram assessment to authenticate reliability.

Motivated by these limitations, the present exploratory effort aims to establish a mathematically grounded CNFM framework, employ similarity transformations to transmute PDEs into ODEs, implement stochastic AI-based LMB neural networks for solution generation, construct a validated dataset using Adam optimization, and rigorously authenticate the algorithm through statistical and graphical validation procedures. By integrating intelligent computing with nonlinear transport theory, this study seeks to enhance predictive capability, computational efficiency, and reliability in analyzing chemically reactive, activation energy-dependent unsteady three-dimensional nanofluid flows.

The paper is organized as follows: Section 2 describes the physical model and governing equations. Section 3 presents the similarity transformations and dimensionless formulation. Section 4 introduces the LMB-ANN methodology. Section 5 discusses the numerical results and performance metrics. Section 6 concludes with major findings and future perspectives.

## 2. Mathematical model and physical assumptions

By assuming the suppositions of said problem are.

- The continuous two-dimensional magnetohydrodynamics stagnation point flow of cross nanoparticles across the melt's enclosing horizontal cylinder of radius  $R$  is taken into account.
- The Darcy-Forchheimer relation is used to assess the flow examination of MHD fluid flow triggered by a permeable material.
- By taking into account the influence of chemical reactions with melting effects and viscous dissipation, solutal analysis and thermal transmission are studied.
- Concentration and temperature at the surface of cylinder  $C_m$  and  $T_m$  correspondingly. Ambient worth of  $\tilde{C}$  and  $\tilde{T}$  as  $r$  tends to infinity are characterized by  $C_\infty$  and  $T_\infty$  correspondingly, where  $T_\infty > T_m$ .
- The stretching cylinder's free stream is provided by  $\tilde{u} \rightarrow U_e(x) = \frac{a_0 x}{l}$  and that of velocity is  $\tilde{u} = U_w = \frac{d_0 x}{l}$ , where  $l$  is characteristics of length and  $a_0$  and  $d_0$  are positive constants.
- The properties of mass and heat diffusion mechanisms are handled through thermophoresis and Brownian motion parameters.

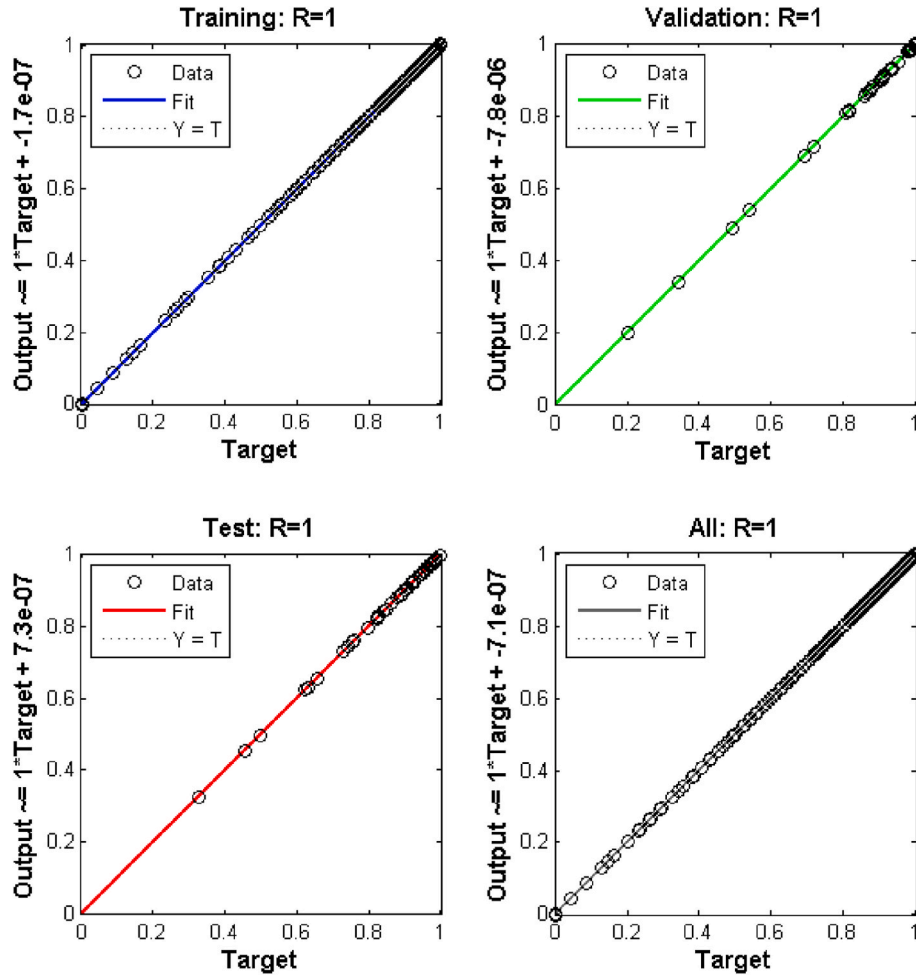


Fig. 11. RA for LMB-Scheme of 3rd snr of CNFM.

Governing equations of flow model are the,

$$\frac{\partial(\tilde{r}\tilde{u})}{\partial x} + \frac{\partial(\tilde{r}\tilde{v})}{\partial r} = 0, \quad (1)$$

$$\frac{\partial(\tilde{r}\tilde{u})}{\partial x} + \frac{\partial(\tilde{r}\tilde{v})}{\partial r} = \frac{1}{\tilde{\rho}} \frac{\partial p}{\partial x} \left[ \frac{\tilde{\frac{\partial \tilde{u}}{\partial r}}}{1 + \left( \Gamma \times \frac{\partial \tilde{u}}{\partial r} \right)^m} \right] + \frac{\mu_0}{\tilde{\rho}} \frac{\partial}{\partial r} \left[ \frac{\tilde{\frac{\partial \tilde{u}}{\partial r}}}{1 + \left( \Gamma \times \frac{\partial \tilde{u}}{\partial r} \right)^m} \right] - \frac{1}{\tilde{\rho}} \times \frac{\mu_0 \tilde{u}}{k^*} - \frac{c_k}{\sqrt{k^*}} \tilde{u}^2 - \frac{\tilde{\sigma}}{\tilde{\rho}} B_0^2 \tilde{u}, \quad (2)$$

Closer to the stagnation area

$$-\frac{1}{\tilde{\rho}} \frac{\partial p}{\partial x} U_e(x) \frac{dU_e(x)}{dx} + \frac{1}{\tilde{\rho}} \frac{\mu_0 U_e(x)}{k^*} + \frac{c_k}{\sqrt{k^*}} U_e^2(x) + \frac{\tilde{\sigma}}{\tilde{\rho}} B_0^2 U_e(x), \quad (3)$$

$$\frac{\partial(\tilde{r}\tilde{u})}{\partial x} + \frac{\partial(\tilde{r}\tilde{v})}{\partial r} = U_e(x) \frac{dU_e(x)}{dx} + \frac{\mu_0}{\tilde{\rho} r} \left[ \frac{\left( \frac{\partial \tilde{u}}{\partial r} \right)}{1 + \left( \Gamma \times \frac{\partial \tilde{u}}{\partial r} \right)^m} \right] + \left( \frac{\mu_0}{\tilde{\rho}} \right) \frac{\partial}{\partial r} \left[ \frac{\frac{\partial \tilde{u}}{\partial r}}{1 + \left( \Gamma \times \frac{\partial \tilde{u}}{\partial r} \right)^m} \right] + \left( \frac{1}{\tilde{\rho}} \right) \frac{\mu_0}{k^*} [U_e(x) - \tilde{u}] + \frac{c_k}{\sqrt{k^*}} [U_e^2(x) - \tilde{u}^2] + \frac{\tilde{\sigma}}{\tilde{\rho}} B_0^2 [U_e(x) - (\tilde{u})], \quad (4)$$

$$\left( \tilde{u} \frac{\partial(\tilde{T})}{\partial x} + (\tilde{v}) \frac{\partial(\tilde{T})}{\partial r} \right) = (\tilde{\alpha}) \left[ \left( \frac{1}{r} \right) \frac{\partial}{\partial r} \left( r \times \frac{\partial \tilde{T}}{\partial r} \right) \right] + \tau \left[ (\tilde{D}_b) \frac{\partial \tilde{T}}{\partial r} \frac{\partial \tilde{C}}{\partial r} + \left( \frac{\tilde{D}_T}{T_\infty} \right) \left( \frac{\partial \tilde{T}}{\partial r} \right)^2 \right] \left( \frac{\mu_0}{\tilde{\rho} c_p^*} \right) \left[ \frac{\left( \frac{\partial \tilde{u}}{\partial r} \right)^2}{1 + \left( \Gamma \times \frac{\partial \tilde{u}}{\partial r} \right)^m} \right], \quad (5)$$

$$\tilde{u} \frac{\partial(\tilde{C})}{\partial x} + \tilde{v} \frac{\partial(\tilde{C})}{\partial r} + \tilde{K}(\tilde{C} - C_\infty) = \tilde{D}_b \left[ \left( \frac{1}{r} \right) \frac{\partial}{\partial r} \left( r \times \frac{\partial \tilde{C}}{\partial r} \right) \right] + \left( \frac{\tilde{D}_T}{T_\infty} \right) \left[ \left( \frac{1}{r} \right) \frac{\partial}{\partial r} \left( r \times \frac{\partial \tilde{T}}{\partial r} \right) \right], \quad (6)$$

Considering the following BCs:

$$\left. \begin{aligned} \text{At } r = R, \tilde{u} = U_w = \frac{d_0 x}{l}, \tilde{T} = T_m, \tilde{C} = C_m, \\ \text{At } r = R, k \frac{\partial \tilde{T}}{\partial r} = \tilde{\rho} [\tilde{\lambda} + c_s(T_m - T_0)] \tilde{v}, \\ \text{As } r \rightarrow \infty, \tilde{u} \rightarrow U_e(x) = \frac{a_0 x}{l}, \tilde{T} \rightarrow T_m, \tilde{C} \rightarrow C_m \end{aligned} \right\} \quad (7)$$

### 3. Similarity transformation and dimensionless equations

By presenting the subsequent appropriate similarity variables as:

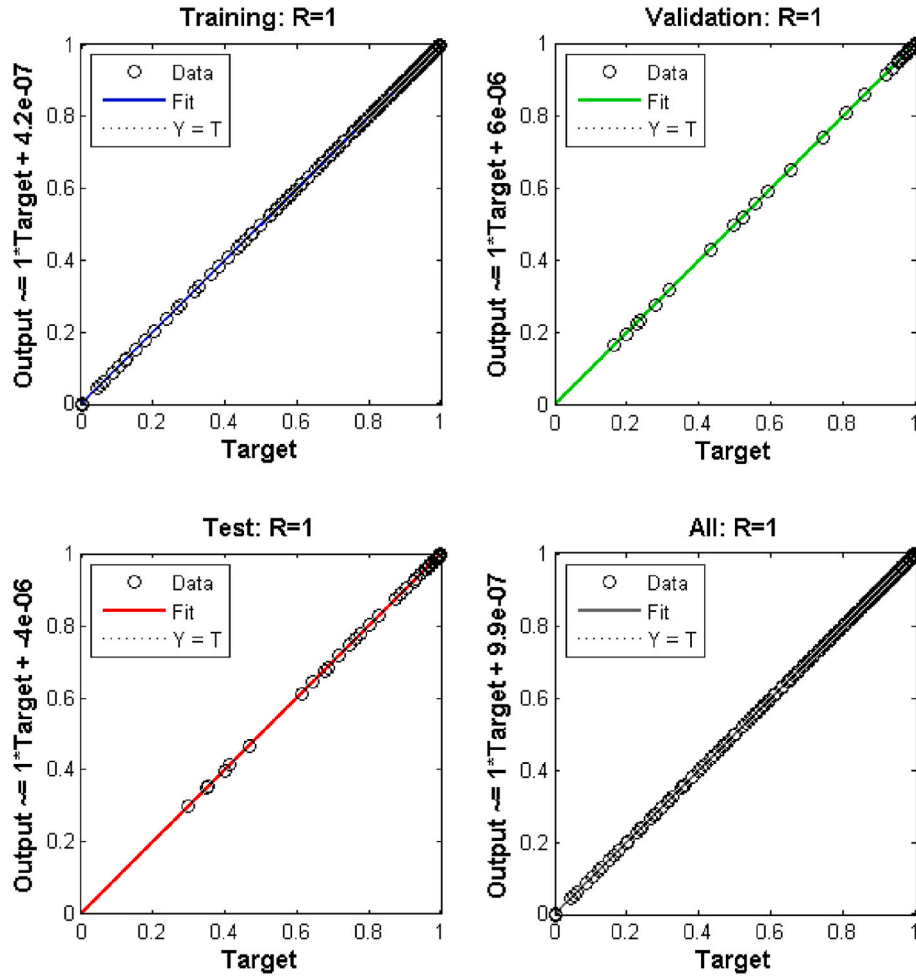


Fig. 12. RA for LMB-Scheme of 4th snr of CNFM.

$$\left. \begin{aligned} \eta &= \sqrt{\frac{d_0}{\nu l}} \left( \frac{r^2 - R^2}{2R} \right), \xi = \sqrt{\frac{\nu d_0 x^2}{l}} f(\eta), \tilde{u} = \frac{1}{r} \frac{\partial \xi}{\partial r}, \tilde{v} = -\frac{1}{r} \frac{\partial \xi}{\partial r}, \\ \theta(\eta) &= \frac{\tilde{T} - T_m}{T_\infty - T_m}, \phi(\eta) = \frac{\tilde{C} - C_m}{C_\infty - C_m}. \end{aligned} \right\} \quad (8)$$

The transformations mentioned in eq. (8) are used in the leading eqs (4)–(6) with boundary conditions in equation (7) to get the ordinary differential equations as:

With BCs:

$$\left. \begin{aligned} f'(\eta) = 1, \theta(\eta) = 0, \phi(\eta) = 0, \Pr f(\eta) + \tilde{M}\theta'(\eta) = 0 \quad \text{at } \eta = 0, \\ f'(\eta) = q, \theta(\eta) = 1, \phi(\eta) = 1 \quad \text{as } \eta \rightarrow \infty. \end{aligned} \right\} \quad (12)$$

The dimensionless parameters are:

$$\begin{aligned} [2\beta(f'') + (1 + 2(\eta)(\beta))f''] \left[ 1 - \left( (f'')\sqrt{(1 + 2(\eta)(\beta))} We \right)^m \right] - m \left( (f'')\sqrt{(1 + 2(\eta)(\beta))} We \right)^m \\ [\beta f'' + (1 + 2(\eta)(\beta))f''] + ff'' - f^2 + w^*(q - f'') + (\tilde{F})(q^2 - f^2) + (\tilde{B}^2)(q^2 - f) + q^2 = 0, \end{aligned} \quad (9)$$

$$\begin{aligned} (1 + 2(\eta)(\beta))(\theta'') + 2\beta(\theta') + \Pr f(\theta') + \Pr \times (1 + 2(\eta)(\beta)) [N_b^*(\theta' \phi') + N_t^*(\theta'^2)] \\ + \Pr Ec f'' (1 + 2(\eta)(\beta)) \left[ 1 - \left( f'' (1 + 2(\eta)(\beta))^{\frac{1}{2}} We \right)^m \right] = 0, \end{aligned} \quad (10)$$

$$(1 + 2(\eta)(\beta))\phi'' + 2\beta\phi' + Sc f \phi' + \frac{N_b^*}{N_b^*} [2\beta\theta' + (1 + 2(\eta)(\beta))\theta''] - N^* Sc \phi = 0. \quad (11)$$

$$\left. \begin{aligned} B^* &= \frac{\tilde{\sigma}}{\rho d_0} B_0^2 l, We = \frac{\Gamma d_0^{3/2} x}{l^{3/2} \sqrt{\nu}}, \beta = \frac{1}{R} \sqrt{\frac{lv}{d_0}}, w^* = \frac{lv}{d_0 k^*}, q = \frac{a_0}{d_0}, \\ \tilde{F} &= \frac{c_k}{\sqrt{k^*}}, \Pr = \frac{\mu_0 c_p^*}{k}, N_t^* = \frac{\tau \tilde{D}_T}{\nu T_\infty} (T_w - T_\infty), N_b^* = \frac{\tau \tilde{D}_b}{\nu} (C_w - C_\infty), \\ Sc &= \frac{\nu}{\tilde{D}_b}, Ec = \left( \frac{d_0 x}{l} \right)^2 \frac{1}{c_p^* (T_w - T_\infty)}, \tilde{M} = \frac{c_p^* (T_w - T_\infty)}{\lambda + c_s (T_m - T_0)}, N^* = \frac{l \tilde{K}}{d_0}. \end{aligned} \right\} \quad (13)$$

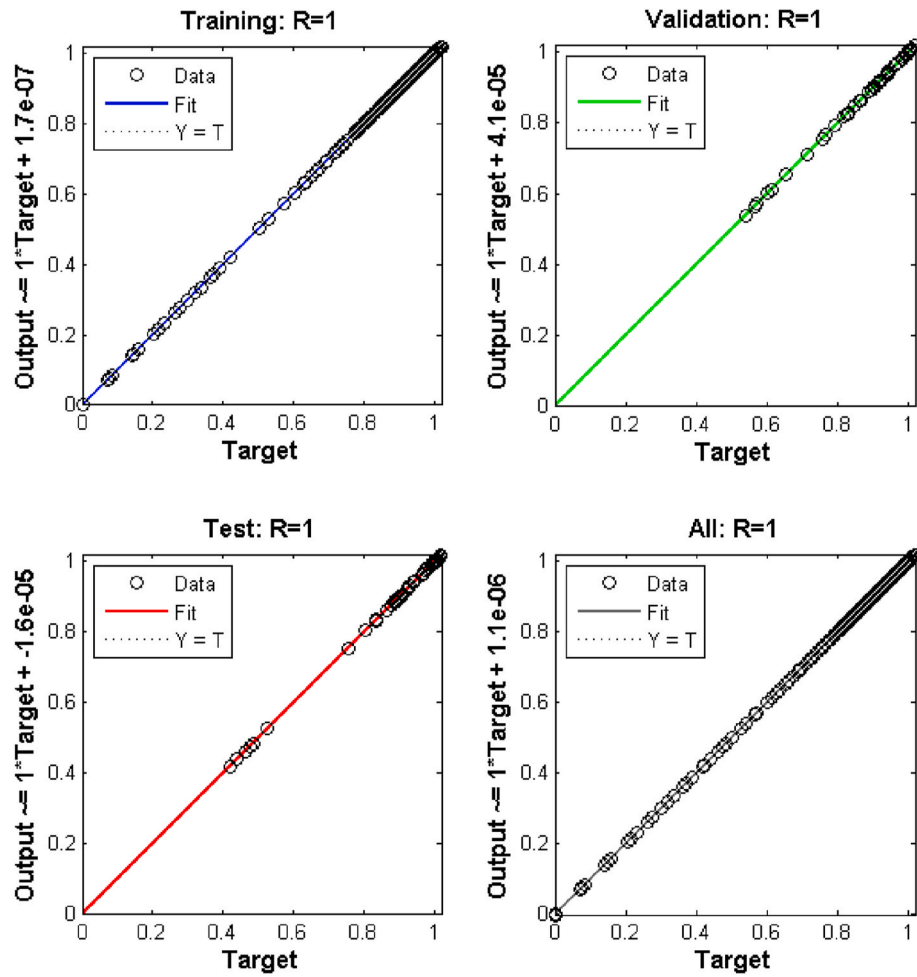


Fig. 13. RA for LMB-Scheme of 5th snr of CNFM.

The expression for Nusselt Number is provided by:

$$\tilde{Nu} = \frac{\tilde{q}_w}{(T_\infty - T_m)}, \text{ where } q_w = -K(\tilde{T}) \frac{\partial(\tilde{T})}{\partial r} \quad (14)$$

With dimensionless physical quantity

$$\tilde{Nu}(\text{Re})_x^{-1/2} = -\theta'(0) \quad (15)$$

The mathematical form for Sherwood Number is provided by:

$$\tilde{Sh}_x = \frac{x\tilde{q}_s}{\tilde{D}_b(\tilde{C} - C_\infty)}, \text{ where } q_s = -\tilde{D}_b \frac{\partial(\tilde{C})}{\partial r} \quad (16)$$

With dimensionless physical quantity

$$\tilde{Sh}_x(\text{Re})_x^{-1/2} = -\phi'(0) \quad (17)$$

Local Skin Friction's mathematical form is provided by:

$$\tilde{C}_f = \frac{\tau_{rx}}{\rho U_w^2}, \text{ where } \tau_{rx} = u_0 \frac{\partial \tilde{u}}{\partial r} \left[ \frac{1}{1 + \left( \Gamma \frac{\partial \tilde{u}}{\partial r} \right)^m} \right] \quad (18)$$

With dimensionless physical quantity

$$\frac{1}{2} \tilde{C}_f(\text{Re})_x^{-1/2} = f''(0) \frac{1}{[1 + ((We)f''(0))^m]} \quad (19)$$

The mathematical form for Local Reynolds is provided by:

$$(\text{Re})_x = \frac{U_w x}{\nu} \quad (20)$$

#### 4. Methodology of solution

By using Table 1 (physical quantities), the Mathematica software's numerical ND-Solve technique, produced the numerical outcomes (see Fig. 1). This numerical dataset contains 101 inputs of concentration, temperature, and velocity profiles are being divided into 3 datasets of validation, testing and training with divisions of (15/100), (15/100), and (71/100) accordingly. This numerical data-set is repeatedly trained and retrained in order to get the lowest MSE values and to justify the accuracy of results. Whereas the NN Structure is shown in Fig. 2, the LMB Algorithm based computing paradigm with a multilayered structure of NNs of AI is employed to mine the requisite data in graphical and statistical form.

Fig. 3 establishes Single Neural Model design for the prophesied NNs of AI of LMB-Scheme. "NFTOOL," an application inside the MATLAB driver's NNs process, is used to represent the AI NNs on LMB Scheme. The LMB Scheme allows for the training of NN weights. Numerical examination is showed for LMB-Scheme, is available here for CNFM given in equations (9)–(11). Table 1 describes the validation of the numerical and computational methodology with an absolute error of about 0.001. The considered LMB-Scheme is applied for all the 6 Snrs by variation of  $q$ ,  $Pr$ ,  $Nb^*$  and  $Sc$  respectively, each with 3 cases of CNFM with fixed values of  $\beta = 3.0$ ,  $We = 0.1$ ,  $m = 1.0$ ,  $w^* = 0.1$ ,  $\tilde{F} = 0.2$ ,  $\tilde{B} = 0.1$ ,  $Nt^* = 0.1$ ,  $Ec = 0.2$ ,  $N^* = 0.1$  and  $\tilde{M} = 1.0$  are tabularized in Table 2. Fig. 4

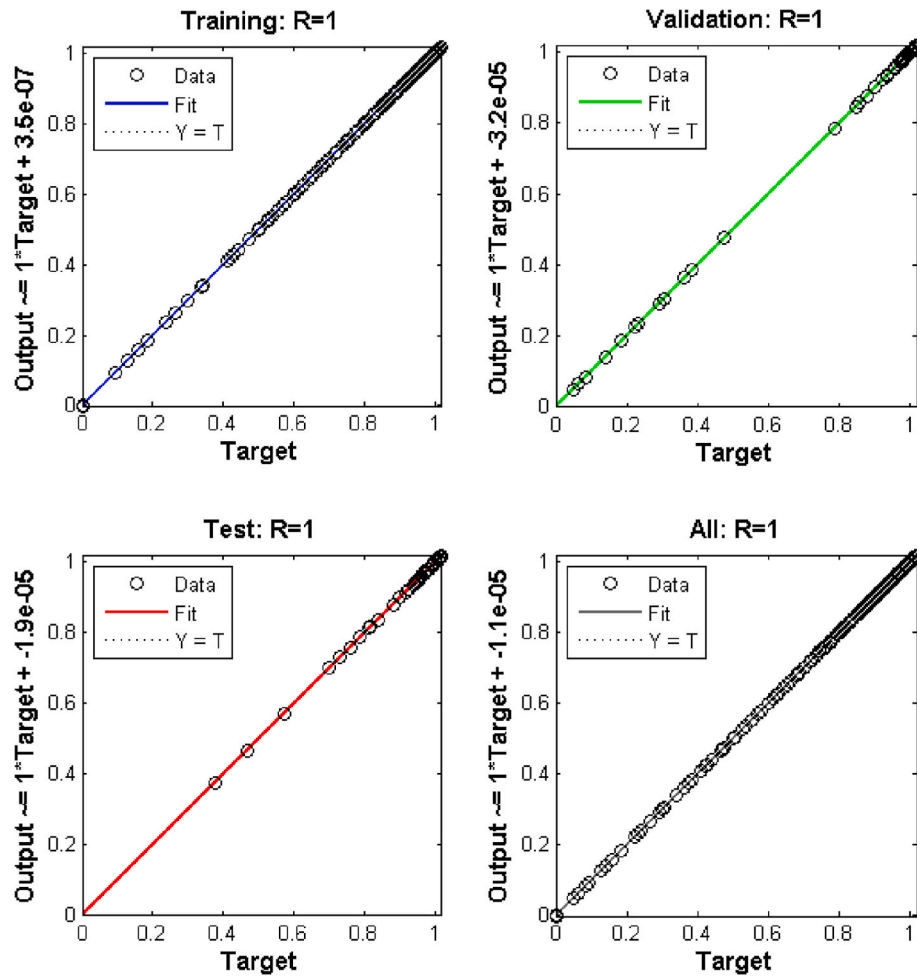


Fig. 14. RA for LMB-Scheme of 6th snr of CNFM.

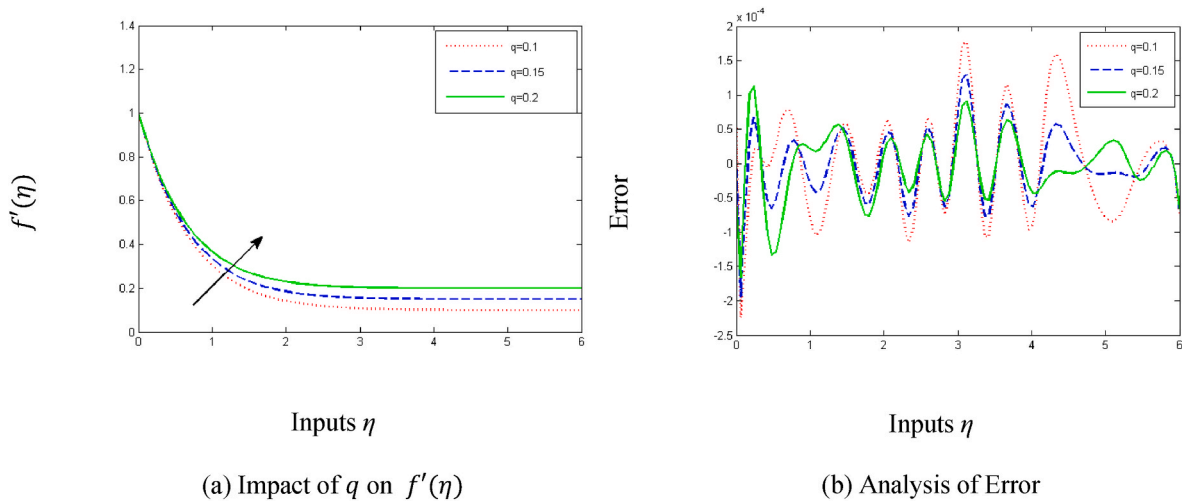


Fig. 15. Assessment of proposed LMB-Scheme with corresponding numerical outcomes for 1st scenario of CNFM.

clarifies the inclusive development of flow map (see Table 3).

## 5. Discussion of outcomes

Fig. 5(a)–(f) displays the MSE Performance Analysis of suggested LMB-Scheme for the solution of CNFM of whole 6 scenarios. At

$5.4272e^{-9}$  alongside 268 Epochs,  $5.3145e^{-9}$  alongside 178 Epochs,  $1.8664e^{-10}$  alongside 753 Epochs,  $3.0778e^{-9}$  alongside 238 Epochs,  $4.4083e^{-10}$  alongside 355 Epochs and  $3.8474e^{-9}$  alongside 296 Epochs for Scenarios (1)–(6) respectively, is the Best Validation performance achieved. Fig. 6(a)–(f) shows the TS of LMB-Scheme for the solution of CNFM, achieving “Val fail” fluctuating from 0 to 1, -1 to 1, 0 to 2, 0 to 1,

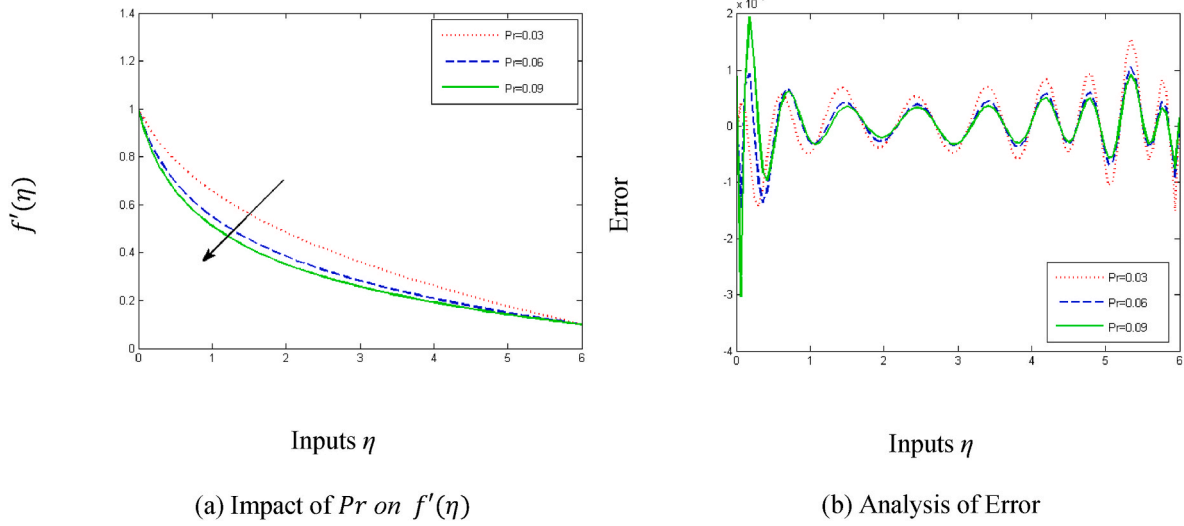


Fig. 16. Assessment of proposed LMB-Scheme with corresponding numerical outcomes for 2nd scenario of CNFM.

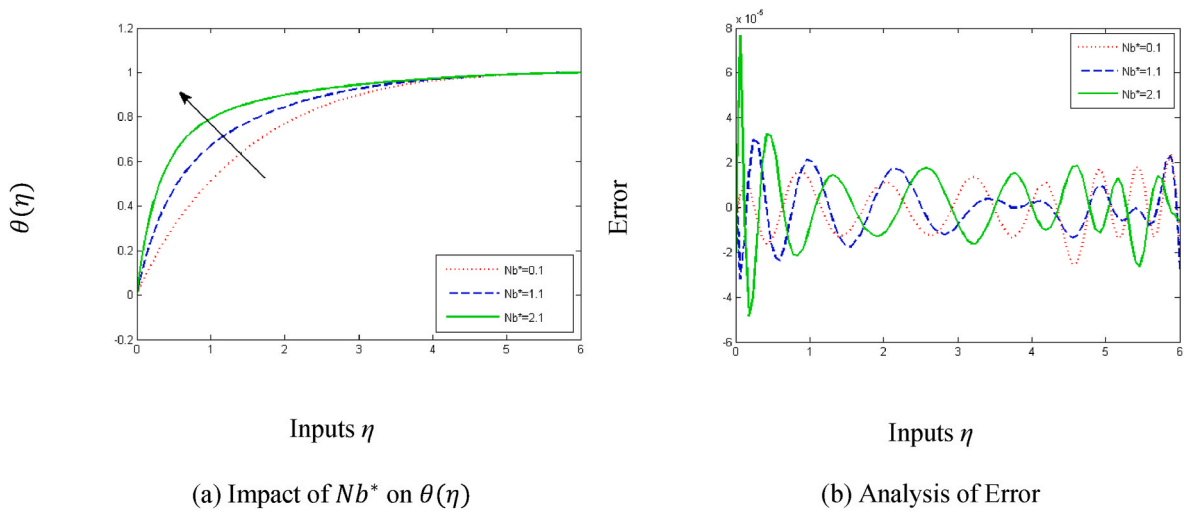


Fig. 17. Assessment of proposed LMB-Scheme with corresponding numerical outcomes for 3rd scenario of CNFM.

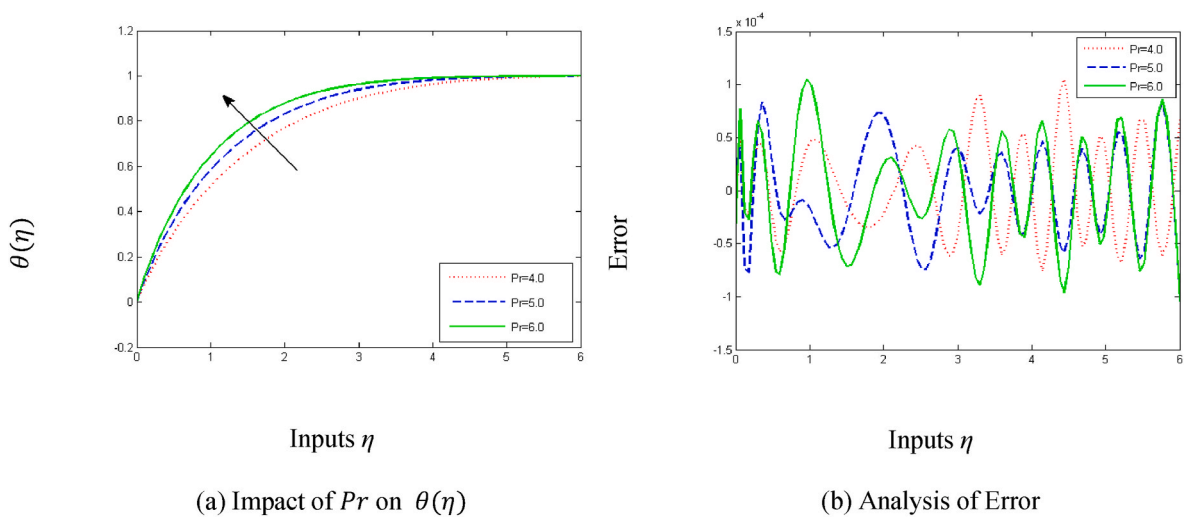


Fig. 18. Assessment of proposed LMB-Scheme with corresponding numerical outcomes for 4th scenario of CNFM.

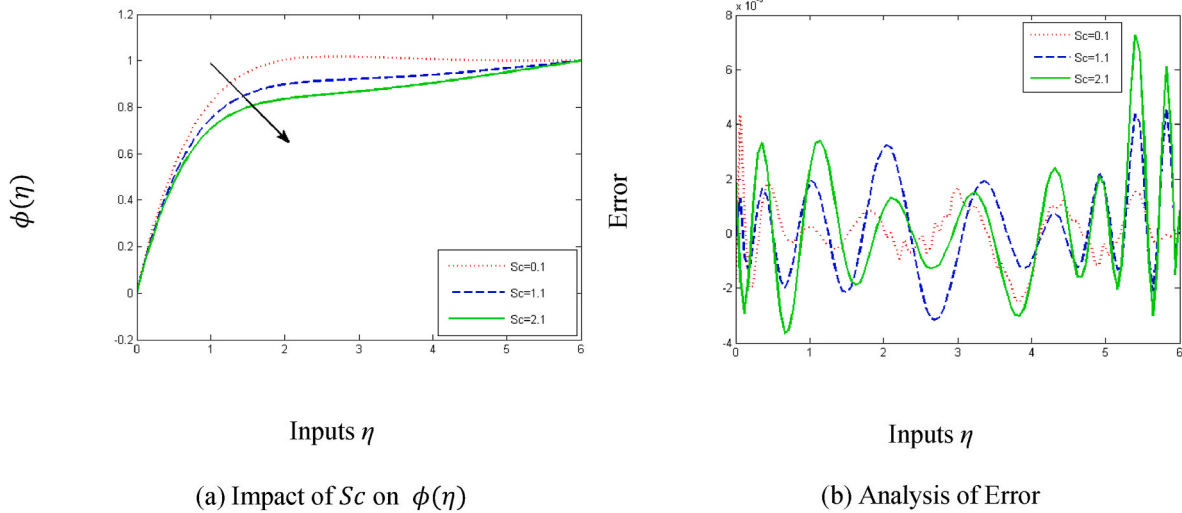


Fig. 19. Assessment of proposed LMB-Scheme with corresponding numerical outcomes for 5th scenario of CNFM.

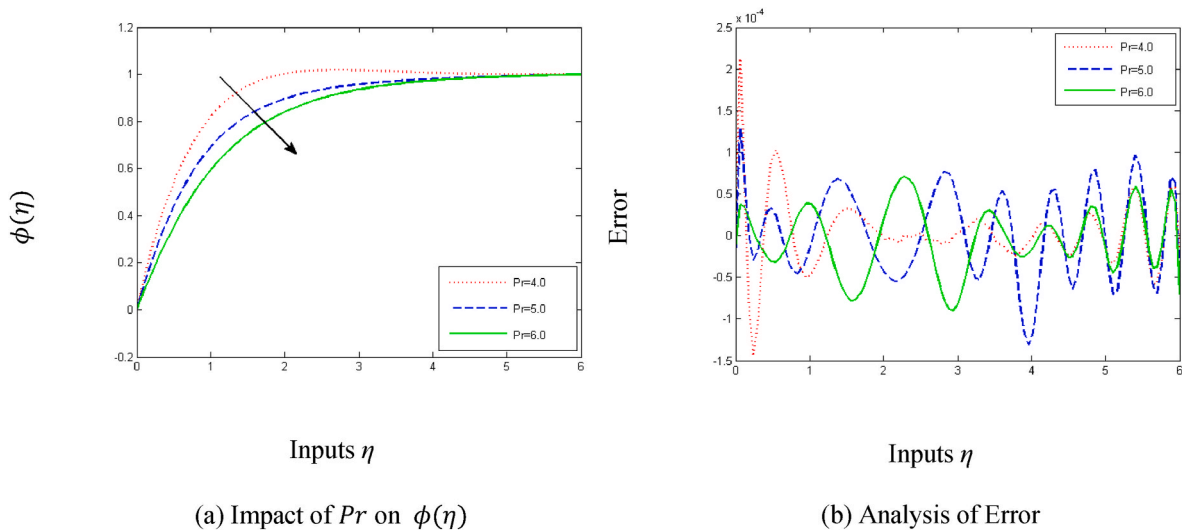


Fig. 20. Assessment of proposed LMB-Scheme with corresponding numerical outcomes for 6th scenario of CNFM.

0 to 2 and -1 to 1 for Scenarios (1)–(6) respectively. For all the six scenarios Fig. 7(a)–(f) particularizes the evaluation of outcomes of FF, achieving the error  $(-2 \text{ to } 2)$ ,  $(-2 \text{ to } 2)$ ,  $(-5 \text{ to } 5)$ ,  $(-2 \text{ to } 2)$ ,  $(-5 \text{ to } 5)$ , and  $(-2 \text{ to } 2)$ . The error is further scrutinized through EH with 20 Bins in Fig. 8(a)–(f) by means of equation  $(\text{Errors} = \text{Targets} - \text{Outputs})$ . The error is shown to be in  $-1.3e^{-5}$  to  $7.22e^{-6}$ ,  $-1.7e^{-5}$  to  $7.68e^{-6}$ ,  $-1.6e^{-6}$  to  $4.65e^{-6}$ ,  $-4.8e^{-6}$  to  $5.64e^{-6}$ ,  $-1.0e^{-6}$  to  $4.46e^{-6}$  and  $-1.0e^{-5}$  to  $7.51e^{-6}$  for Scenarios (1)–(6) respectively. For all the scenarios Figs. 9–14 illustrates the plots of RA for LMB-Scheme effects of CNFM at  $(R = 1)$ . For validation, testing and training procedures, a researcher could investigate the enhanced convergence performance gained at Epochs 268, 178, 753, 238, 355 and 296 alongside performance grids  $2.77 \text{ E}^{-9}$ ,  $1.68 \text{ E}^{-9}$ ,  $1.85 \text{ E}^{-10}$ ,  $1.89 \text{ E}^{-9}$ ,  $2.14 \text{ E}^{-10}$  and  $1.57 \text{ E}^{-9}$  correspondingly as displayed in MSE Table. Parallel gradient grids values are  $9.78 \text{ E}^{-8}$ ,  $9.81 \text{ E}^{-8}$ ,  $9.89 \text{ E}^{-9}$ ,  $9.91 \text{ E}^{-8}$ ,  $9.98 \text{ E}^{-8}$  and  $9.94 \text{ E}^{-8}$  and  $Mu$  Grids are  $10^{-9}$  to  $10^{-8}$ . Consequently, for temperature, concentration, and velocity profiles, results of LMB Scheme are attained. Effects of  $f'(\eta)$  are shown in Fig. 15 (a) for Scenario 1 and 16(a) for Scenario 2. Effects of  $\theta(\eta)$  are visualized in Fig. 17(a) for Scenario 3 and 18(a) for Scenario 4. Outcomes of  $\phi(\eta)$  are visualized in Fig. 19(a) for Scenario 5 and 20(a) for Scenario 6. Fig. 15(a) displays increasing behavior of  $f'(\eta)$  for  $q$  and 16(a) displays the declining pattern of  $f'(\eta)$  for  $Pr$ . Velocity increases with stretching

ratio due to enhanced momentum diffusion near the wall. Velocity decreases with higher Prandtl number because of reduced thermal diffusivity. Temperature increases with Brownian motion and thermophoresis, indicating nanoparticle-induced energy transport. Concentration decreases with higher Schmidt number due to reduced mass diffusivity. Fig. 17(a) and 18(a) displays the increasing behavior of  $\theta(\eta)$  for  $Nb^*$  and  $Pr$  respectively. Also, Fig. 19(a) and 20(a) demonstrate the declining pattern of concentration profile  $\phi(\eta)$  for  $Sc$  and  $Pr$ . Accordingly, Fig. 15(b)–20(b) demonstrates the Error Analysis for all the scenarios. By incorporating AI technologies like LMB-Scheme into the MATLAB chamber, complete numerical results are obtained. Additionally, AI metrics are now interpreted in physical terms, low MSE implies reliable heat and mass transfer predictions under varying physical constraints. The validity, correctness, and convergence were confirmed by comparing the ensuing graphical productivities with the reference data.

## 6. Conclusions and future scope

This study presented an intelligent computational framework for analyzing the unsteady magnetohydrodynamic (MHD) Carreau nano-fluid model (CNFM) over a stretchable surface under the effects of

thermal radiation and variable viscosity. By employing similarity transformations, the governing nonlinear PDE system was reduced to ODEs and solved using a hybrid approach that integrates MATLAB-based Artificial Neural Networks (ANNs) trained with the Levenberg–Marquardt backpropagation (LMB) scheme alongside the conventional ND-Solve numerical solver for validation. The proposed model systematically examined the influence of key physical parameters, including magnetic parameter, thermal conductivity, Prandtl number, heat source parameter, curvature parameter, Schmidt number, and Brownian motion parameter, on velocity, temperature, and concentration distributions across multiple physical scenarios.

The results demonstrate that the machine learning (ML)-assisted LMB neural network provides highly accurate approximations, achieving mean squared error (MSE) values in the range of  $10^{-10}$  to  $10^{-09}$ , thereby confirming strong convergence and predictive capability. The robustness of the solver was further validated through regression analysis, error histograms (EH), convergence curves, and performance analysis across training (71%), validation (15%), and testing (15%) datasets using a 10-neuron hidden layer architecture. Compared with purely traditional solvers, the ML-enhanced framework offers improved computational efficiency, reduced dependence on iterative discretization schemes, enhanced generalization capability, and reliable parametric sensitivity analysis. These advantages highlight the value-added contribution of integrating machine learning techniques with nonlinear non-Newtonian MHD nanofluid modeling, particularly for complex multi-parameter systems where classical methods may become computationally intensive.

From the physical analysis, the principal findings include.

- An increasing trend in the velocity profile  $f'(\eta)$  with increasing curvature parameter  $q$ , while higher Prandtl number reduces velocity distribution;
- An enhancement in the temperature profile  $\theta(\eta)$  with increasing Brownian motion parameter  $Nb^*$  and Prandtl number;
- A decreasing concentration profile  $\phi(\eta)$  with increasing Schmidt number and Prandtl number;
- Consistent validation of all six modeled scenarios through numerical and graphical agreement.
- Overall, the study confirms that LMB-based neural networks can serve as a reliable and efficient intelligent solver for nonlinear CNFM systems.

### 6.1. Future research directions

Future investigations may extend this framework by.

- Incorporating physics-informed neural networks (PINNs) to directly embed governing equations into the loss function for enhanced physical consistency;
  - Coupling experimental or real-world nanofluid data with data-driven ML models to strengthen predictive validation;
  - Exploring deeper or adaptive neural architectures to improve scalability for higher-dimensional and multiphysics problems;
  - Applying the methodology to transient industrial processes involving reactive nanofluids, biomedical flows, and energy transport systems.
- By bridging advanced machine learning techniques with nonlinear fluid mechanics, this work establishes a foundation for next-generation intelligent solvers in complex transport phenomena.

### CRediT authorship contribution statement

**Imran Abbasi:** Conceptualization, Visualization, Investigation, Writing - original draft, Writing - review & editing. **Mushtaq K.**

**Abdallah:** Conceptualization, Visualization, Investigation, Writing - original draft, Writing - review & editing. **Waqar Azeem Khan:** Conceptualization, Visualization, Investigation, Writing - original draft, Writing - review & editing. **Zohaib Arshad:** Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Masood Khan:** Conceptualization, Visualization, Investigation, Writing - original draft, Writing - review & editing. **Emad Ghandourah:** Software, editing, Funding Acquisition, project management, Methodology, Visualization, Formal analysis, Writing - original draft, Supervision. **Ali Morfeq:** Software, editing, Funding acquisition, Project management, Methodology, Visualization, Formal analysis, Writing - original draft, Supervision. **Faisal J. Alzahrani:** Software, editing, Funding Acquisition, project management, Methodology, Visualization, Formal analysis, Writing - original draft, Supervision.

### Declaration of competing interest

I have no conflict of interest.

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### Data availability

No data was used for the research described in the article.

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